

Introduction to Flash Memory

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Invited Paper

The most relevant phenomenon of this past decade in the field of semiconductor memories has been the explosive growth of the Flash memory market, driven by cellular phones and other types of electronic portable equipment (palm top, mobile PC, mp3 audio player, digital camera, and so on). Moreover, in the coming years, portable systems will demand even more nonvolatile memories, either with high density and very high writing throughput for data storage application or with fast random access for code execution in place. The strong consolidated know-how (more than ten years of experience), the flexibility, and the cost make the Flash memory a largely utilized, well-consolidated, and mature technology for most of the nonvolatile memory applications. Today, Flash sales represent a considerable amount of the overall semiconductor market.

Although in the past different types of Flash cells and architectures have been proposed, today two of them can be considered as industry standard: the common ground NOR Flash, that due to its versatility is addressing both the code and data storage segments, and the NAND Flash, optimized for the data storage market.

This paper will mainly focus on the development of the NOR Flash memory technology, with the aim of describing both the basic functionality of the memory cell used so far and the main cell architecture consolidated today. The NOR cell is basically a floating-gate MOS transistor, programmed by channel hot electron and erased by Fowler–Nordheim tunneling. The main reliability issues, such as charge retention and endurance, will be discussed, together with the understanding of the basic physical mechanisms responsible. Most of these considerations are also valid for the NAND cell, since it is based on the same concept of floating-gate MOS transistor.

Furthermore, an insight into the multilevel approach, where two bits are stored in the same cell, will be presented. In fact, the exploitation of the multilevel approach at each technology node allows the increase of the memory efficiency, almost doubling the density at the same chip size, enlarging the application range, and reducing the cost per bit.

Finally, the NOR Flash cell scaling issues will be covered, pointing out the main challenges. The Flash cell scaling has been demonstrated to be really possible and to be able to follow the Moore's law down to the 130-nm technology generations. The technology development and the consolidated know-how is expected to sustain the scaling trend down to the 90- and 65-nm technology nodes as forecasted by the International Technology Roadmap of Semiconductors. One of the crucial issues to be solved

to allow cell scaling below the 65-nm node is the tunnel oxide thickness reduction, as tunnel thinning is limited by intrinsic and extrinsic mechanisms.

Keywords—Flash evolution, Flash memory, Flash technology, floating-gate MOSFET, multilevel, nonvolatile memory, NOR cell, scaling.

I. INTRODUCTION

The semiconductor market, for the long term, has been continuously increasing, even if with some valleys and peaks, and this growing trend is expected to continue in the coming years (see Fig. 1). A large amount of this market, about 20%, is given by the semiconductor memories, which are divided into the following two branches, both based on the complementary metal–oxide–semiconductor (CMOS) technology (see Fig. 2).

- The volatile memories, like SRAM or DRAM, that although very fast in writing and reading (SRAM) or very dense (DRAM), lose the data contents when the power supply is turned off.
- The nonvolatile memories, like EPROM, EEPROM, or Flash, that are able to balance the less-aggressive (with respect to SRAM and DRAM) programming and reading performances with nonvolatility, i.e., with the capability to keep the data content even without power supply.

Thanks to this characteristic, the nonvolatile memories offer the system a different opportunity and cover a wide range of applications, from consumer and automotive to computer and communication (see Fig. 3).

The different nonvolatile memory families can be qualitatively compared in terms of flexibility and cost (see Fig. 4). Flexibility means the possibility to be programmed and erased many times on the system with minimum granularity (whole chip, page, byte, bit); cost means process complexity and in particular silicon occupancy, i.e., density or, in simpler words, cell size. Considering the flexibility-cost plane, it turns out that Flash offers the best compromise between these two parameters, since they have the smallest cell size

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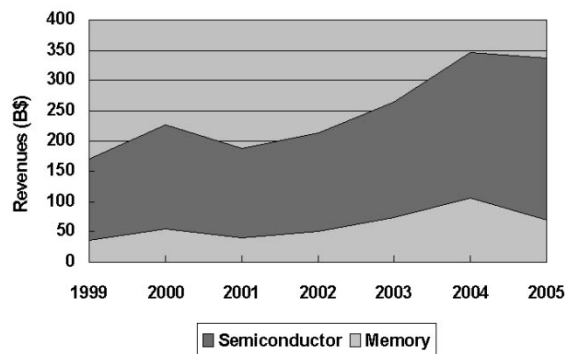


Fig. 1. Semiconductor market: revenues versus year. The bottom wave refers to the semiconductor memory amount.

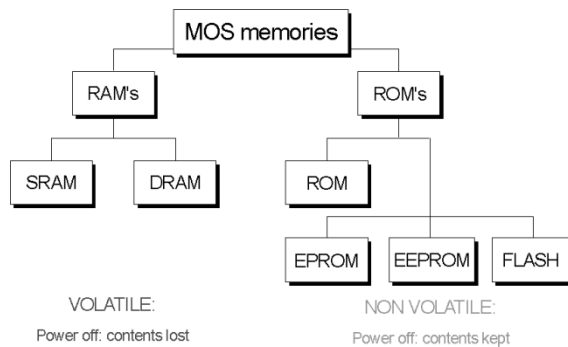


Fig. 2. MOS memory tree.

	Cell Phone	Consumer	Automotive	Computer & Communication
EPROM	Analog, Residential	Games, Set Top Box	Engine Mgt.	HDD, Copiers, Fax, Switching
FLASH	Digital (GSM)	Set Top Box, PDA	All Power Train, Car Navigation, ABS, GPS	HDD, PC Bios, CDROM
EEPROM	Digital (GSM)	Audio, Video	All Car Body	PC SPD, Graphic boards, Printers

Fig. 3. Main nonvolatile memory applications.

(one transistor cell) with a very good flexibility (they can be electrically written on field more than 100 000 times, with byte programming and sectors erasing).

The most relevant phenomenon of this past decade in the field of semiconductor memories has been the explosive growth of the Flash memory market, driven by cellular phones and other types of electronic portable equipment (palm top, mobile PC, mp3 audio player, digital camera, and so on). Moreover, in the coming years, portable systems will demand even more nonvolatile memories, either with high density and very high writing throughput for data storage application or with fast random access for code execution in place.

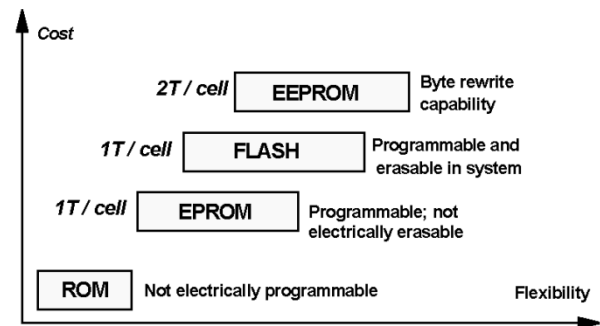


Fig. 4. Nonvolatile memory (NVM) qualitative comparison in the flexibility–cost plane. A common feature of NVMs is to retain the data even without power supply.

Based on these market needs, a well-known way to classify Flash products and the relative technologies is that of defining two major application segments:

- code storage, where the program or the operating system is stored and is executed by the microprocessor or microcontroller;
- data (or mass) storage, where data files for image, music, and voice are recorded and read sequentially.

Different type of Flash cells and architectures have been proposed in the past (see Fig. 5). They can be divided in terms of access type, parallel or serial, and in terms of the utilized programming and erasing mechanism, Fowler–Nordheim tunneling (FN), channel hot electron (CHE), hot-holes (HH), and source-side hot electron (SSHE). Among all of these architectures, today two can be considered as industry standard: the common ground NOR Flash [1]–[3], that due to its versatility is addressing both the code and data storage segments, and the NAND Flash, optimized for the data storage market [4], [5].

In the following, the basic concepts, the reliability issues, the evolution, and scaling trends will be presented only for the NOR Flash cell, but most of these considerations are also valid for the NAND since both of them are based on the concept of floating-gate MOS transistor.

II. NOR FLASH CELL

In 1971, Frohman-Bentchkowsky presented a floating gate transistor in which hot electrons were injected and stored [6], [7]. From this original work, the erasable programmable read only memory (EPROM) cell, programmed by CHE and erased by ultraviolet (UV) photoemission, has been developed. The EPROM technology became the most important nonvolatile memory in the 1980s. In the same period, the Flash EEPROM was proposed, basically an EPROM cell, with the possibility to be electrically erased [8]. The name Flash was given to represent the fact that the whole memory array could be erased in the same (fast) time.

The first Flash product was presented in 1988 [9]. In terms of applications, initially Flash products were mainly used as an “EPROM replacement,” offering the possibility to be erased on system, avoiding the cumbersome UV erase oper-

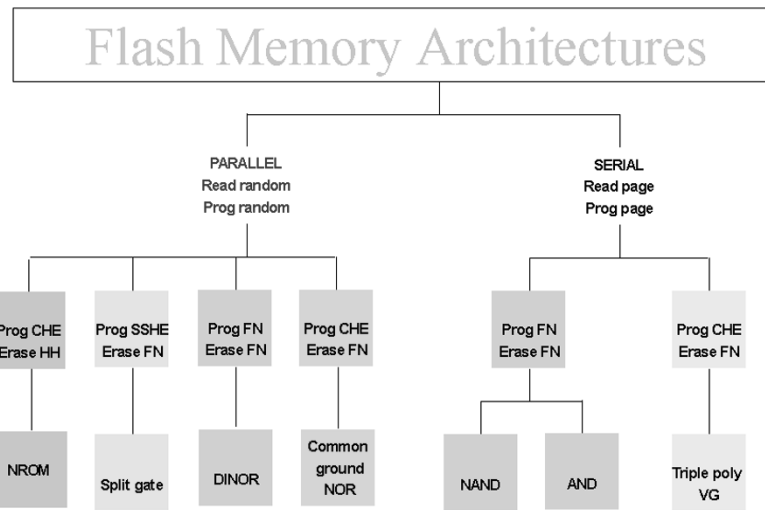


Fig. 5. The family tree of Flash memory cell architecture. The actual industry standard are: 1) The NOR for code and data storage application and 2) NAND only for data storage.

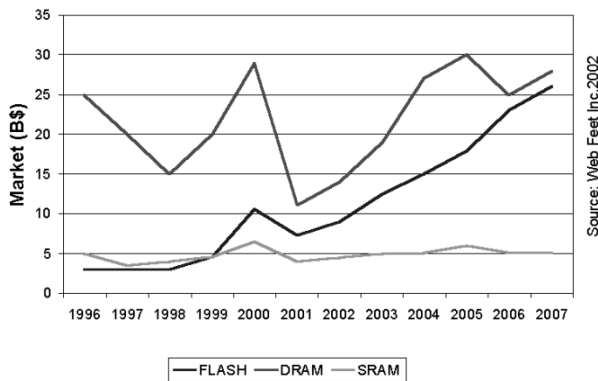


Fig. 6. Semiconductor memory market for the main memory, i.e., DRAM, Flash, and SRAM.

ation. But the Flash market did not take off until this technology was proven to be reliable and manufacturable. In the late 1990s, the Flash technology exploded as the right non-volatile memory for code and data storage, mainly for mobile applications. Starting from 2000, the Flash memory can be considered a really mature technology: more than 800 million units of 16-Mb equivalent NOR Flash devices were sold in that year.

In Fig. 6, the Flash market is reported and compared with the DRAM and SRAM one [10]. It can be seen that the Flash market became and has stayed bigger than the SRAM one since 1999. Moreover, the Flash market is forecasted to be above \$20 billion in three or four years from now, reaching the DRAM market amount, and only smoothly following the DRAM oscillating trend, driven by the personal computer market. In fact, portable systems for communications and consumer markets, which are the drivers of the Flash market, are forecasted to continuously grow in the coming years.

In the following, we briefly describe the basics of the Flash cell functionality.

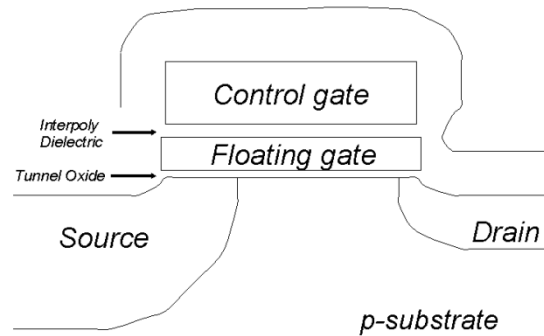


Fig. 7. Schematic cross section of a Flash cell. The floating-gate structure is common to all the nonvolatile memory cells based on the floating-gate MOS transistor.

A. Basic Concept

A Flash cell is basically a floating-gate MOS transistor (see Fig. 7), i.e., a transistor with a gate completely surrounded by dielectrics, the floating gate (FG), and electrically governed by a capacitively coupled control gate (CG). Being electrically isolated, the FG acts as the storing electrode for the cell device; charge injected in the FG is maintained there, allowing modulation of the “apparent” threshold voltage (i.e., V_T seen from the CG) of the cell transistor. Obviously the quality of the dielectrics guarantees the non-volatility, while the thickness allows the possibility to program or erase the cell by electrical pulses. Usually the gate dielectric, i.e., the one between the transistor channel and the FG, is an oxide in the range of 9–10 nm and is called “tunnel oxide” since FN electron tunneling occurs through it. The dielectric that separates the FG from the CG is formed by a triple layer of oxide–nitride–oxide (ONO). The ONO thickness is in the range of 15–20 nm of equivalent oxide thickness. The ONO layer as interpoly dielectric has been introduced in order to improve the tunnel oxide quality. In fact, the

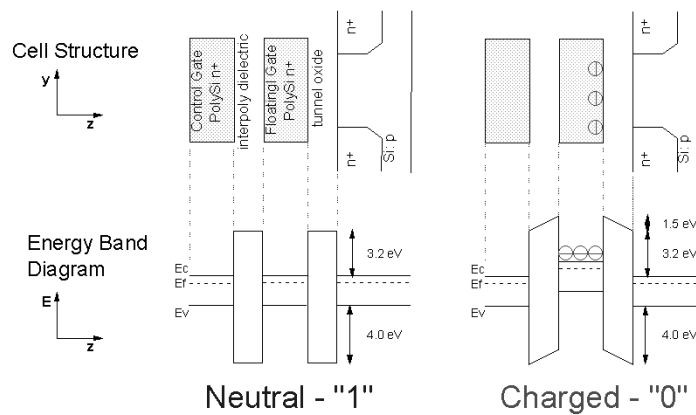


Fig. 8. Schematic energy band diagram (lower part) as referred to a floating gate MOSFET structure (upper part). The left side of the figure is related to a neutral cell, while the right side to a negatively charged cell.

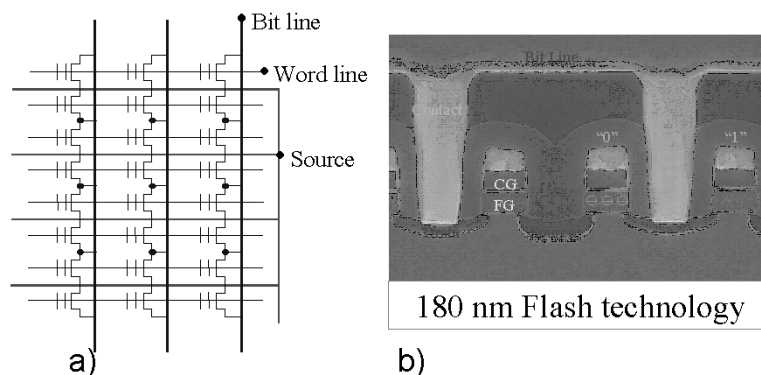


Fig. 9. (a) NOR Flash array equivalent circuit. (b) Flash memory cell cross section.

use of thermal oxide over polysilicon implies growth temperature higher than 1100 °C, impacting the underneath tunnel oxide. High-temperature postannealing is known to damage the thin oxide quality.

If the tunnel oxide and the ONO behave as ideal dielectrics, then it is possible to schematically represent the energy band diagram of the FG MOS transistor as reported in Fig. 8. It can be seen that the FG acts as a potential well for the charge. Once the charge is in the FG, the tunnel and ONO dielectrics form potential barriers.

The neutral (or positively charged) state is associated with the logical state “1” and the negatively charged state, corresponding to electrons stored in the FG, is associated with the logical “0.”

The “NOR” Flash name is related to the way the cells are arranged in an array, through rows and columns in a NOR-like structure. Flash cells sharing the same gate constitute the so-called wordline (WL), while those sharing the same drain electrode (one contact common to two cells) constitute the bitline (BL). In this array organization, the source electrode is common to all of the cells [Fig. 9(a)].

A scanning electron microscope (SEM) cross section along a bitline of a Flash array is reported in Fig. 9(b), where three cells can be observed, sharing two by two the drain

contact and the sourceline. This picture can be better understood considering the layout of a cell (see Fig. 10) and the two schematic cross sections, along the y direction (bitline) and the x direction (wordline). The cell area is given by the x pitch times the y pitch. The x pitch is given by the active area width and space, considering also that the FG must overlap the oxide field. The y pitch is constituted by the cell gate length, the contact-to-gate distance, half contact, and half sourceline. It is evident, as reported in Fig. 9(b), that both contact and sourceline are shared between two adjacent cells.

B. Reading Operation

The data stored in a Flash cell can be determined measuring the threshold voltage of the FG MOS transistor. The best and fastest way to do that is by reading the current driven by the cell at a fixed gate bias. In fact, as schematically reported in Fig. 11, in the current–voltage plane two cells, respectively, logic “1” and “0” exhibit the same transconductance curve but are shifted by a quantity—the threshold voltage shift (ΔV_T)—that is proportional to the stored electron charge Q .

Hence, once a proper charge amount and a corresponding ΔV_T is defined, it is possible to fix a reading voltage in such

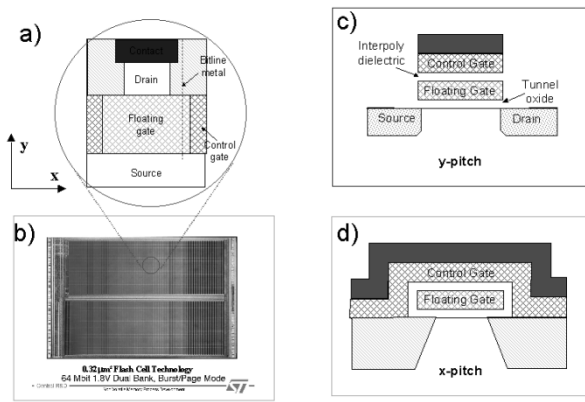


Fig. 10. The NOR Flash cell. (a) Basic layout. (b) Updated Flash product (64-Mb, 1.8-V Dual bank). (c) and (d) are, respectively, the schematic cross section along bitline (y pitch) and wordline (x pitch).

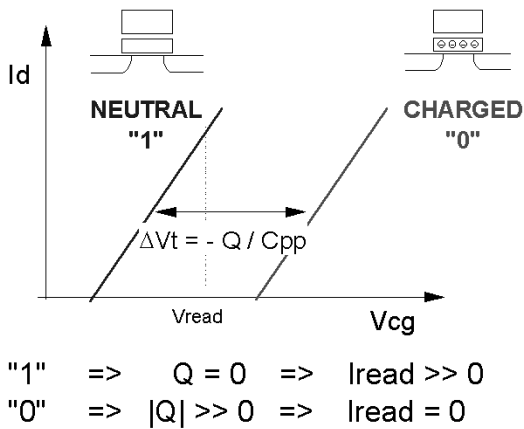


Fig. 11. Floating-gate MOSFET reading operation.

a way that the current of the “1” cell is very high (in the range of tens of microamperes), while the current of the “0” cell is zero, in the microampere scale. In this way, it is possible to define the logical state “1” from a microscopic point of view as no electron charge (or positive charge) stored in the FG and from a macroscopic point of view as large reading current. Vice versa, the logical state “0” is defined, respectively, by electron charge stored in the FG and zero reading current.

C. Writing Operation

Considering Fig. 8, the problem of writing an FG cell corresponds to the physical problem of forcing an electron above or across an energy barrier. The problem can be solved exploiting different physical effects [11]. In Fig. 12, the three main physical mechanisms used to write an FG memory cell are sketched.

- The CHE mechanism, where electrons gain enough energy to pass the oxide–silicon energy barrier, thanks to the electric field in the transistor channel between source and drain. In fact, the electron energy distribution presents a tail in the high energy side that can be modulated by the longitudinal electric field.

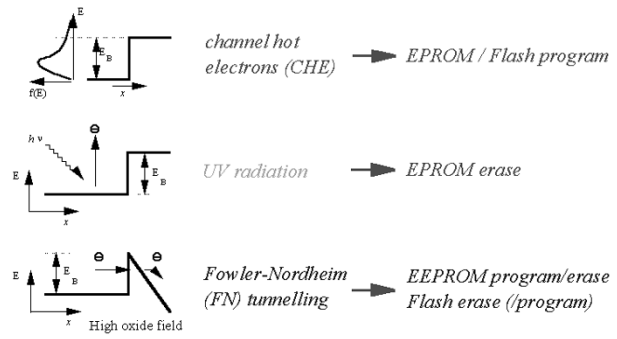


Fig. 12. Writing mechanism in floating-gate devices.

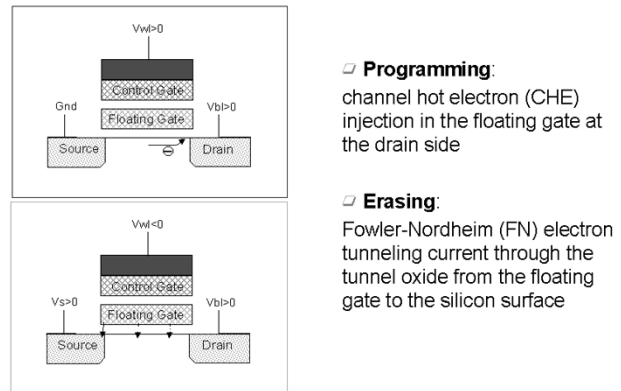


Fig. 13. NOR Flash writing mechanism.

- The photoelectric effect, where electrons gain enough energy to surmount the barrier thanks to the interaction with a photon with energy $h\nu$ larger than the barrier itself. For silicon–dioxide, this corresponds to UV radiation. This mechanism is the one originally used in EPROM’s products to erase the entire device.
- The Fowler–Nordheim electron tunneling mechanism is a quantum-mechanical tunnel induced by an electric field. Applying a strong electric field (in the range of 8–10 MV/cm) across a thin oxide, it is possible to force a large electron tunneling current through it without destroying its dielectric properties.

A NOR Flash memory cell is programmed by CHE injection in the FG at the drain side and it is erased by means of the FN electron tunneling through the tunnel oxide from the FG to the silicon surface (see Fig. 13).

III. RELIABILITY

Many issues have to be addressed when, from the theoretical model of a single cell, a Flash product has to be realized, integrating millions of cells in an array. Nonvolatility implies at least ten years of charge retention, and the data must be stored in a cell after many read/program/erase cycles. The confidence in Flash memory reliability has grown together with the understanding of the single memory cell failure mechanisms.

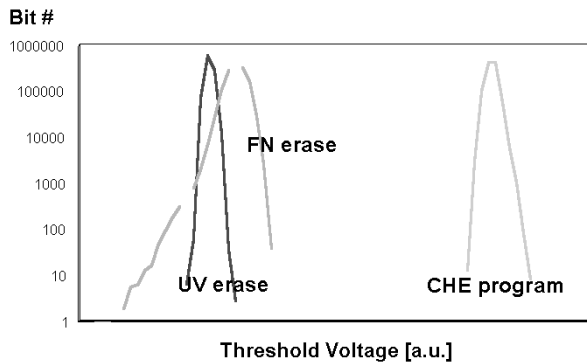


Fig. 14. Threshold voltage distribution of a 1-Mb Flash array after UV erasure, after CHE programming, and after FN erasure.

The high degree of testability [12] allows the detection at wafer level of latent defects which may cause single-cell failures related to programming disturbs, data retention, and oxide defects [13], thus making Flash one of the most reliable nonvolatile memories.

A. Threshold Voltage Distribution

When dealing with a large array of cells, e.g., from tens of thousands to one million, it is very important to understand the type of dispersion given by the large set of cells. The best way to do it is to compare the threshold voltage distribution of the whole array, considering it after UV erasure—that can be considered as the reference state—after CHE programming and after FN erasing.

Fig. 14 shows typical distributions of cell threshold voltages in a large memory array. The UV-erased distribution is pretty narrow and symmetrical. A more accurate analysis would reveal a Gaussian distribution due to random variations of critical dimensions, thickness, and doping which contribute to cause a dispersion of threshold voltages, either directly or through coupling ratios.

The programmed distribution is wider than the UV-erased one, but it is still symmetrical. The enlargement occurs because most of the parameters that cause V_T dispersion of UV-erased cells also impact the threshold shift of programmed cells.

The distribution of threshold voltages after electrical erase is much wider and heavily asymmetrical. A more detailed analysis would show that the bulk of the distribution is again a Gaussian with a standard deviation larger than the one of programmed cells. Cells in this part of the distribution are referred to as “normal” cells. But there is also an exponential tail at low V_T , composed of cells that erase faster than the average, also called “tail” cells.

The dispersion of threshold voltages of normal cells is due to coupling ratio variations, and it has been accurately modeled [14]. Instead, the understanding of the tail cells, although of key importance, is more difficult. In fact, as these cells erase faster than normal cells with the same applied voltage, one should assume that they are somehow “defective.” However, they are just too numerous for being associated with extrinsic defects.

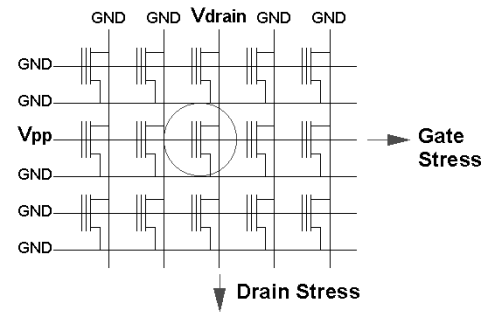


Fig. 15. Schematic of a Flash array, showing row and column disturbs occurring when the cycled cell is programmed.

Different models have been presented with the aim to explain the tail cells. For example, a distribution in the polycrystalline structure of the FG, with a barrier height variation at the grain boundaries, would give rise to a local enhancement of the tunnel barrier [15]. Another model explains the tail cells as due to randomly distributed positive charges in the tunnel oxide [16]. This model is solidly based on the well-known existence of donor-like bulk oxide traps and on calculations that show the huge increase of the tunnel current density caused by the presence of an elementary positive charge closed to injecting electrode.

Independently from a consolidated model, it can be stated that the exponential tail of the erased distribution is mostly related to structural imperfections, i.e., intrinsic defects, and it can be minimized by process optimization (for example, working on silicon surface preparation, tunnel oxidation, FG polysilicon optimization) but not eliminated. Flash products must be designed taking into account the existence of such a tail.

B. Program Disturb

The failure mechanisms referred to as “program disturbs” concern data corruption of written cells caused by the electrical stress applied to these cells while programming other cells in the memory array. Two types of program disturbs must be taken into account: row and column disturbs, also referred as gate and drain stress, as schematically reported in Fig. 15, representing a portion of a cell array.

Row disturbs are due to gate stress applied to a cell while programming other cells on the same wordline. If a high voltage is applied to the selected row, all the other cells of that row must withstand the gate stress without losing their data. Depending on the data stored in the cells, data can be lost either by a leakage in the gate oxide or by a leakage in the interpoly dielectric.

Column disturbs are due to drain stress applied to a cell while programming other cells on the same bitline. Under this condition, programmed cells can lose charge by FN tunneling from the FG to the drain (soft erasing). The program disturb depends on the number of cells along bitline and wordline and then depends strongly on the sector organization. The most effective way to prevent disturb propagation is to use block select transistor in a divided bitline and wordline

organization to completely isolate each sector. Program disturb really could be a critical issue in Flash memory, and cells and circuits must be designed with safety margins versus the stress sensitivity.

C. Data Retention

As in any nonvolatile memory technology, Flash memories are specified to retain data for over ten years. This means the loss of charge stored in the FG must be as minimal as possible. In updated Flash technology, due to the small cell size, the capacitance is very small and at an operative programmed threshold shift—about 2 V—corresponds a number of electrons in the order of 10^3 to 10^4 . A loss of 20% in this number (around 2–20 electrons lost per month) can lead to a wrong read of the cell and then to a data loss.

Possible causes of charge loss are: 1) defects in the tunnel oxide; 2) defects in the interpoly dielectric; 3) mobile ion contamination; and 4) detrapping of charge from insulating layers surrounding the FG.

The generation of defects in the tunnel oxide can be divided into an extrinsic and an intrinsic one. The former is due to defects in the device structure; the latter to the physical mechanisms that are used to program and erase the cell. The tunnel oxidation technology as well as the Flash cell architecture is a key factor for mastering a reliable Flash technology.

The best interpoly dielectric considering both intrinsic properties and process integration issues has been demonstrated to be a triple layer composed of ONO. For several generations, all Flash technologies have used ONO as their interpoly dielectric.

The problem of mobile ion contamination has been already solved on the EPROM technology, taking particular care with the process control, but in particular using high phosphorus content in intermediate dielectric as a gettering element. [17], [18]. The process control and the intermediate dielectric technology have also been implemented in the Flash process, obtaining the same good results.

Electrons can be trapped in the insulating layers surrounding the floating gate during wafer processing, as a result of the so-called plasma damage, or even during the UV exposure normally used to bring the cell in a well-defined state at the end of the process. The electrons can subsequently detrapp with time, especially at high temperature. The charge variation results in a variation of the floating gate potential and thus in cell V_T decrease, even if no leakage has actually occurred. This apparent charge loss disappears if the process ends with a thermal treatment able to remove the trapped charge.

The retention capability of Flash memories has to be checked by using accelerated tests that usually adopt screening electric fields and hostile environments at high temperature.

D. Programming/Erasing Endurance

Flash products are specified for 10^5 erase/program cycles. Cycling is known to cause a fairly uniform wear-out of the cell performance, mainly due to tunnel oxide degradation, which eventually limits the endurance characteristics [19]. A

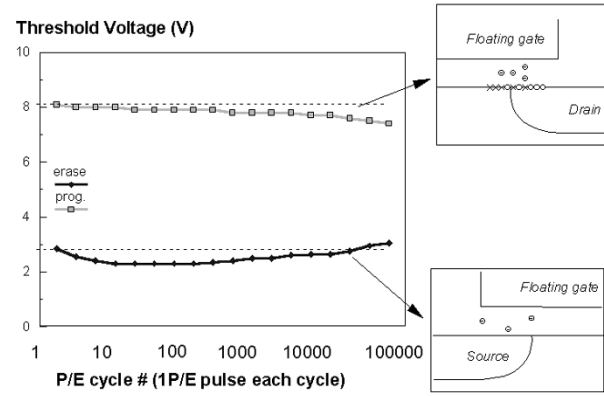


Fig. 16. Threshold voltage window closure as a function of program/erase cycles on a single cell.

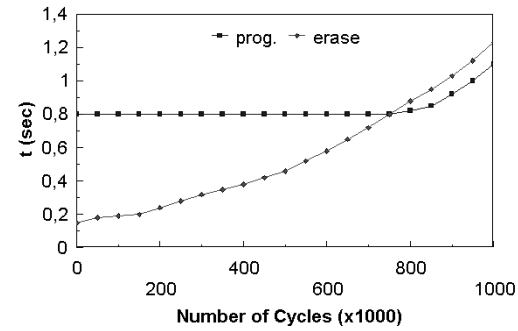


Fig. 17. Program and erase time as a function of the cycles number.

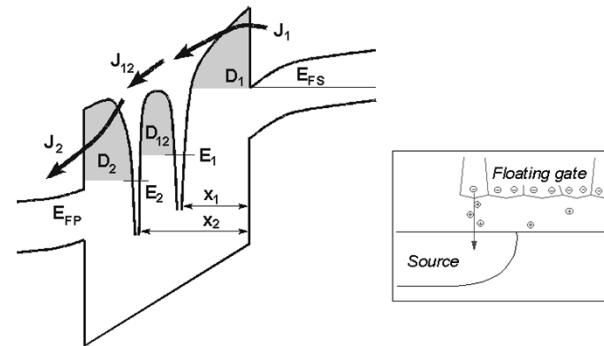


Fig. 18. Anomalous SILC modeling. The leakage is caused by a cluster of positive charge generated in the oxide during erase (left-hand side). The multitrapp assisted tunneling is used to model SILC: trap parameters are energy and position.

typical result of an endurance test on a single cell is shown in Fig. 16. As the experiment was performed applying constant pulses, the variations of program and erase threshold voltage levels are described as “program/erase threshold voltage window closure” and give a measure of the tunnel oxide aging. In real Flash devices, where intelligent algorithms are used to prevent V_T window closing, this effect corresponds to a program and erase times increase (see Fig. 17).

In particular, the reduction of the programmed threshold with cycling is due to trap generation in the oxide and to interface state generation at the drain side of the channel, which are mechanisms specific to hot-electron degradation.

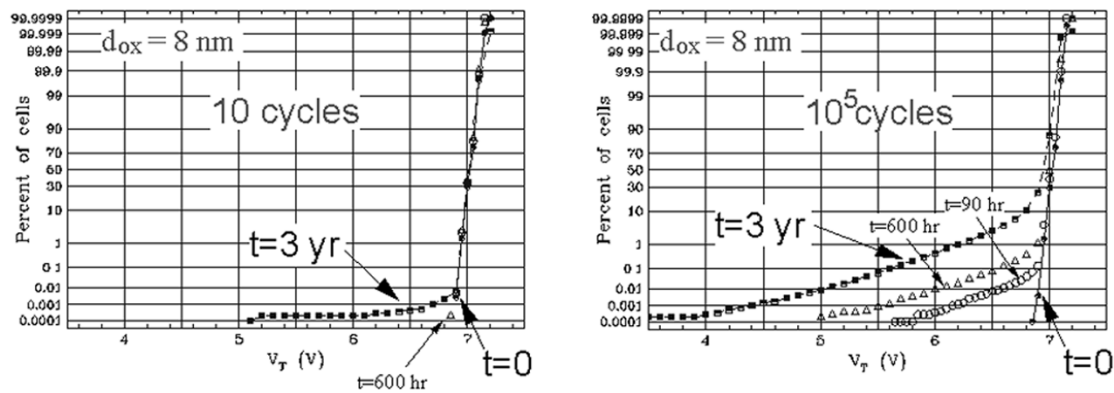


Fig. 19. Data retention tests at room temperature.

The evolution of the erase threshold voltage reflects the dynamics of net fixed charge in the tunnel oxide as a function of the injected charge. The initial lowering of the erase V_T is due to a pile-up of positive charge which enhances tunneling efficiency, while the long-term increase of the erase V_T is due to a generation of negative traps.

Cycling wear-out can be reduced by proper device engineering and by optimization of the tunnel oxide process. However, once process and product are qualified for a given endurance specification, no major problems should come from lot-to-lot variation.

Actually, endurance problems are mostly given by single-cell failures, which present themselves like a retention problem after program/erase cycles. In fact, a high field stress on thin oxide is known to increase the current density at low electric field. The excess current component, which causes a significant deviation from the current-voltage curves from the theoretical FN characteristics at low field, is known as stress-induced leakage current (SILC). SILC is clearly attributed to stress-induced oxide defects and, as far as a conduction mechanism, it is attributed to a trap assisted tunneling (see Fig. 18). The main parameters controlling SILC are the stress field, the amount of charge injected during the stress, and the oxide thickness. For fixed stress conditions, the leakage current increases strongly with decreasing oxide thickness [20]–[22].

The effect of cycling on data retention cannot be referred to in the typical cell, but must be studied considering a wide array of cells, looking in particular to the tail distribution. In Fig. 19, we report the results of retention test on a 1-Mb array of cells with 8-nm tunnel oxide in order to enhance the SILC defects in single cells. Retention tests have been performed on arrays cycled 10 and 10^5 times [23]. As can be seen, the amount of cells that lose charge after three years are much more in the case of longer endurance. Data retention after cycling is the issue that definitely limits the tunnel oxide thickness scaling. For very thin oxide, below 8–9 nm, the number of leaky cells becomes so large that even error-correction techniques cannot fix the problem.

Flash Memory Scaling

Al Fazio

Abstract

In order to meet technology scaling in the field of solid-state memory and data storage, the mainstream transistor-based flash technologies will start evolving to incorporate material and structural innovations. Dielectric scaling in nonvolatile memories is approaching the point where new approaches will be required to meet the scaling requirements while simultaneously meeting the reliability and performance requirements of future products. High-dielectric-constant materials are being explored as possible candidates to replace the traditional SiO_2 and ONO (oxide/nitride/oxide) films used today in memory cells. Likewise, planar-based memory cell scaling is approaching the point where scaling constraints force exploration of new materials and nonplanar, three-dimensional scaling alternatives. This article will review the current status and discuss the approaches being explored to provide scaling solutions for future transistor floating-gate-based nonvolatile memory products. Based on the introduction of material innovations, it is expected that the planar transistor-based flash memory cells can scale through at least the end of the decade (2010) using techniques that are available today or projected to be available in the near future. More complex, structural innovations will be required to achieve further scaling.

Keywords: flash memory, floating gates, nonvolatile memory, scaling.

Introduction

Floating gate flash memory^{1,2} is now the fastest-growing memory segment, driven by the rapid growth of portable devices such as digital cameras and cellular phones. The technology allows for data stored in multiple memory cells to be erased in a single action (a “flash”) by means of an applied voltage. Flash memory is categorized into two basic approaches: NOR flash, characterized by a fast initial access time for high read performance; and NAND flash, characterized by a slow initial access time and high write performance. This article will focus on the scaling of NOR flash memory; however, many of the scaling considerations described in this article are common to both NOR and NAND floating-gate flash memories.

As flash memory devices begin to scale into the sub-100-nm lithography regime, scaling is becoming a greater challenge due to the high electric fields required for the programming and erase operations and the stringent leakage requirements for long-term charge storage. These requirements are imposing fundamental scaling limitations on the cell operating voltages and on the physical thickness of the tunneling dielectric. Overcoming these limitations will require innovations in cell structure and device materials. This article will outline the

major scaling challenges for flash memory and identify some of the potential solutions to enable further scaling to occur.

Flash Cell Basic Operation

A floating gate nonvolatile flash memory cell is shown schematically in Figure 1. It is a metal oxide semiconductor (MOS) transistor with two gates, a floating gate and a control gate. The memory cell consists of an n -channel transistor with the addition of an electrically isolated polysilicon floating gate. Electrical access to the floating gate is only through a capacitor network of surrounding SiO_2 layers and source,

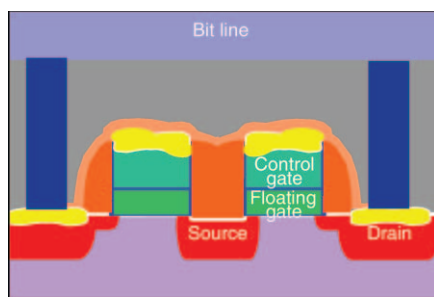


Figure 1. Flash memory cell: cross section along the channel.

drain, transistor channel, and polysilicon control gate terminals. Any charge present on the floating gate is retained due to the inherent Si-SiO₂ energy barrier height, leading to the nonvolatile nature of the memory cell. Characteristic of the structure is a thin tunneling oxide (~100 Å), an oxide/nitride/oxide (ONO) interpoly dielectric (IPD) that resides between the two polysilicon gates, and a short electrical channel length. Because the only electrical connection to the floating gate is through capacitors, the flash cell can be thought of as a linear capacitor network with an n -channel transistor attached. The threshold voltage of the device can be changed by modifying the charge on the floating gate, which can retain this charge for many years. Data can be stored in the memory by adding or removing charge. In the simplest form of a memory cell, two threshold levels (high and low) can store one bit of information in each cell. This concept can be extended to store more levels, commonly called MLC (multilevel cell).³ Four levels would allow two bits of data per cell to be stored. Adding and removing charge from the floating gate is normally achieved by Fowler–Nordheim tunneling or hot carrier injection.

Flash Cell Programming

Programming a flash cell means that charge, or electrons, are added to the floating gate. Figure 2 shows the cell bias conditions during program operation. A high drain-to-source bias voltage is applied, along with a high control gate voltage. The gate voltage inverts the channel (turns it to the ON state), while the drain bias accelerates electrons toward the drain. Programming a flash cell by channeling electrons with high kinetic energy—so-called hot electrons—can be understood by use of the “lucky” electron model,⁴ as illustrated by the energy-band diagram in Figure 3. In the model, an electron crosses the channel without collision—a process that requires some luck—thereby gaining sufficient kinetic energy to surmount the 3.2 eV Si-SiO₂ energy barrier. Prior to entering the drain and being swept away, this lucky electron experiences a collision with the

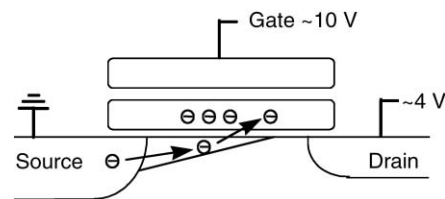


Figure 2. Memory cell bias conditions for programming.

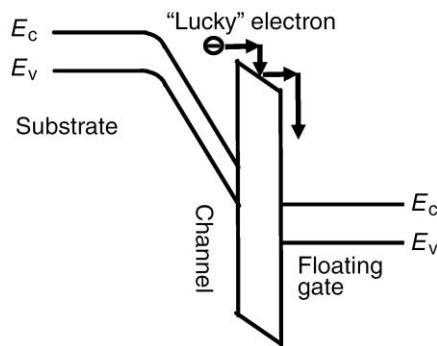


Figure 3. Energy-band diagram for channel hot-electron programming. E_c is the energy of the conducting band and E_v is the energy of the valence band.

silicon lattice and is redirected toward the Si-SiO₂ interface with the aid of the gate field. The electron is subsequently captured on the floating gate and retained as stored charge.

Erasing a Flash Cell

As mentioned earlier, in a flash cell, the contents of the memory, or charge, are removed by means of applying an electrical voltage, erasing them in a "flash." The electrical erasure of flash memory is achieved by Fowler-Nordheim tunneling,⁵ for which the bias conditions are shown in Figure 4. Under these conditions, a high field (8–10 MV/cm) is present between the floating gate and the channel. As evidenced by the energy-band diagram of Figure 5, electrons tunneling through the first ~30 Å of the SiO₂ are then swept into the silicon. When the erase operation has been completed, electrons have been removed from the floating gate, reducing the cell threshold. While programming is selective to each individual cell, erasing is not, with many cells (typically, 64 kbytes) being erased simultaneously.

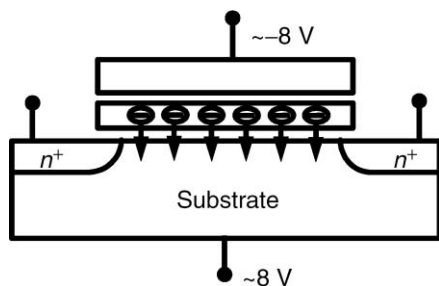


Figure 4. Memory cell bias conditions for erase operation. The floating gate is the rectangle in the middle, and the channel is the region between the n^+ areas at the top of the substrate layer.

Flash Memory Scaling Challenges and Alternatives

Figure 6 depicts the layout of a flash memory cell and some of the major scaling constraints. Throughout the history of the development of flash, the cell size has been reduced through a combination of lithography and self-alignment techniques.⁶ Today, it is common for many of the cell components to be fully or partially self-aligned, eliminating the need for lithography registration between layers. Figure 7 illustrates various self-alignment techniques that have been deployed. Once the cell becomes fully self-aligned, further scaling is largely determined by the electrical limiters of the cell, where scaling the flash cell requires scaling the cell's capacitive network and basic transistor features. The following sections explore how the introduction of new materials and structures will enable flash memory to continue scaling beyond simple linear extrapolations.

Channel-Length Scaling: Challenges and Alternatives

To achieve hot carrier programming, a voltage of about 4 V is required from the drain to source to produce electrons of high enough energy to overcome the 3.2 eV Si-SiO₂ barrier height. One can extrapolate the historical scaling trend for channel length and extrapolate potential limits. This extrapolation is based on the fact that the basic planar cell structure is the same for all of the generations, and scaling is achieved by reducing specific cell dimensions. The difference between the gate length L_{gate} and the effective electrical gate length L_{eff} is the lateral diffusion of the source and drain underneath the gate. The convergence point of the trend represents a projection of a scaling rate limiter of the current planar cell structure. As shown in Figure 8, scaling the gate length will be limited below the 70 nm lithography node due to the inability of the shorter channel length to withstand the required programming voltage. To continue scaling at the same rate, more revolutionary ideas will be needed to scale the L_{eff} more aggressively.

Scaling the channel length could be achieved through engineering of the Si-SiO₂ barrier. By choosing dielectric alternatives to SiO₂, the barrier can be tailored to allow hot-electron injection to occur at lower voltages. Some promising alternatives have been proposed.⁷ Three-dimensional structures⁸ are an alternative way to address the gate-length scaling constraint. Both "fin" (protruding above the silicon like a fish's fin) and "U" (a U-shaped trough etched below the silicon surface) structures show promise to allow further scaling while maintaining the channel length required for pro-

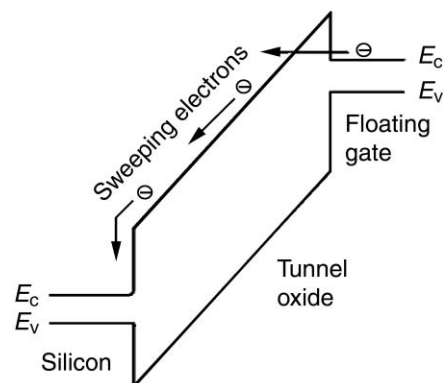


Figure 5. Energy-band diagram for Fowler-Nordheim tunneling erase operation.

gramming. Figure 9 shows an example of a "finfet" (a MOS field-effect transistor made in a fin structure) type of flash cell with a vertical storage gate. This structure moves the channel-length constraint into the vertical (z) direction, allowing further x - y scaling to occur.

Capacitor Network Scaling Challenges and Alternatives

Maintaining adequate coupling of the control gate to the floating gate is a key aspect in scaling the cell channel. This is required to keep the erase voltage as low as possible, avoid erase saturation, and provide control of the channel for reading and programming. Erase saturation occurs when the field across the interpoly dielectric is equal to the field in the tunnel oxide, so that for each electron that is removed (erased) from the floating gate, another electron is added through the IPD. As the cell scales and self-aligned techniques are used for the floating gate, maintaining control of the channel requires a thinner IPD between the control gate and the floating gate, approaching the thickness of the tunneling dielectric. The requirement of an IPD is to provide good capacitive coupling to the floating gate from the control gate, while minimizing any leakage through the dielectric. Unlike the tunnel oxide, this dielectric stack is not expected to support charge transfer during programming and erase operations. For the last decade, the IPD has been composed of an ONO sandwich film. ONO scaling has been achieved by thinning the dielectrics while maintaining control of the thickness. This scaling has also been enabled by improving the quality, in terms of the number of imperfections, of the oxide on the floating gate, which can be aided by reducing the floating-gate polysilicon surface roughness.

An alternative approach would be to replace the IPD with a higher-dielectric-constant material as compared to SiO_2 . To maintain the high capacitive coupling improvement, a metal gate would be required to reduce the series capacitance from gate depletion. Since this film is not expected to transport charge (as is the case with the tunnel oxide), it has fewer constraints and it may be easier to find a candidate to meet the cell requirements, as compared with replacing the tunnel oxide. Possible op-

tions include Al_2O_3 with a TaN gate that has been proposed for a SONOS-type memory device,⁹ that could in principle be deployed in a floating-gate device as well. However, the gate electrode should be chosen consistent with the high- κ film to make sure the energy barrier is engineered to reduce carrier injection.

Another scaling limitation of a capacitor network is the communication of two adjacent cells through the capacitance between the cells. As the spacing between

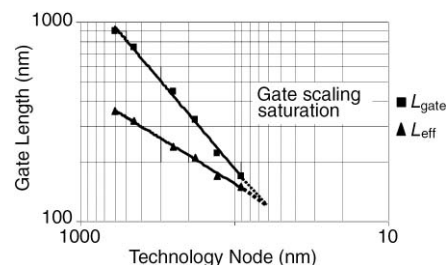


Figure 8. Gate-length scaling projection. L_{gate} is the gate length; L_{eff} is the effective electrical gate length.

floating gates is reduced, coupling between floating gates increases; thus, the data stored in one cell can influence the operation of an adjacent cell (Figure 10). Communication between adjacent cells may be reduced through a reduction in the thickness of the floating gate and control gate, thus reducing the area of the capacitor. Alternative nonconducting floating gate structures are being actively investigated in the industry as ways to replace the normally conducting floating gate with nonconducting floating nodes. The two main thrusts in this area are silicon nitride storage¹⁰ and nanocrystal storage, including metal-based and semiconductor nanocrystals.¹¹ While these approaches offer the possibility of replacing the floating gate, both require significant changes in processing, cell operation (program, erase, read) and structure in order to achieve a viable memory. As such, these approaches face significant barriers to becoming mainstream alternatives to floating gate storage.

Tunnel Oxide Scaling Challenges and Alternatives

Scaling of the tunneling oxide is a primary limitation to cell scaling. The tunnel

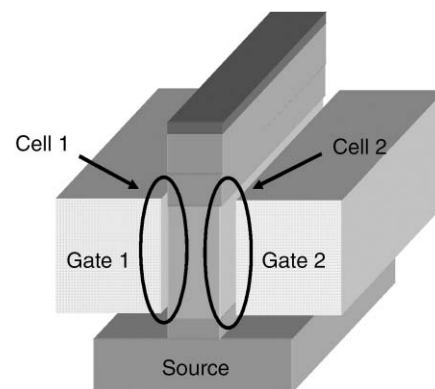


Figure 9. Three-dimensional "finfet" (fin-shaped field-effect transistor) flash structure with a vertical storage transistor.

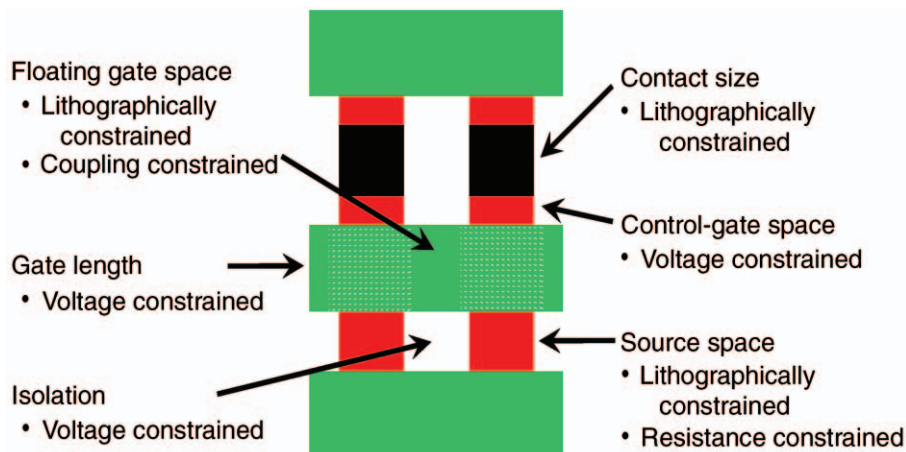


Figure 6. Flash cell layout and major scaling constraints. The lighter-green areas are flash cell gates.

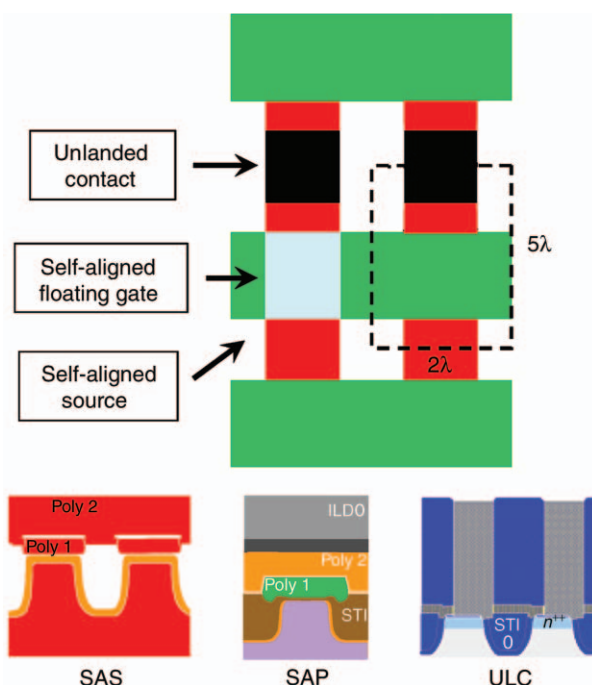


Figure 7. Self-alignment techniques. Use of a self-aligned source (SAS), self-aligned poly (SAP) floating gate, and unlabeled contact (ULC) eliminates most lithography registration components of the flash cell area. The dotted box represents one unit memory cell within an array. λ is the minimum lithography feature size; ILD = interlayer dielectric; STI = shallow trench isolation.

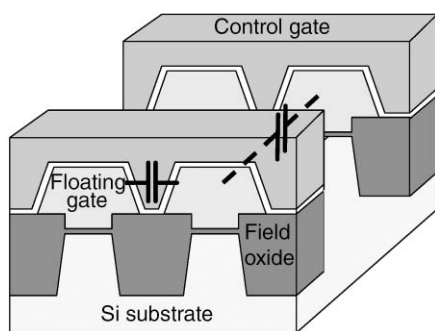


Figure 10. Capacitive coupling between neighboring cells.

oxide needs to scale to enable channel-length scaling, but this is not possible, due to data retention requirements. The charge retention requirements set a lower limit for tunnel oxide thickness of about 80 Å, due to stress-induced low-field tunneling,¹² as shown in Figure 11. In fact, the International Technology Roadmap for Semiconductors (ITRS) revealed a tunnel oxide scaling roadmap that was made more conservative in the 2001 edition due to the challenges in going below 80 Å. It is becoming clear that to enable a significant reduction below 80 Å, a significant change in the SiO₂ dielectric is required. Alternative materials are being explored to allow further scaling of the tunneling dielectric. In particular, "crested barrier"^{13,14} composite films may have promise in enabling the design of a more optimal barrier structure. With SiO₂, or any material with a uniform barrier, the high transparency at electric fields necessary for fast program and erase operations (~10 MV/cm) cannot be achieved along with low transparency at fields (1–3 MV/cm) where good retention is required. The reason for this is that the barrier height remains fixed regardless of the applied electric field, and field emission occurs due to thinning of the barrier width (Figure 12a). If a triangular or crested barrier approach is used (Figure 12b), then the

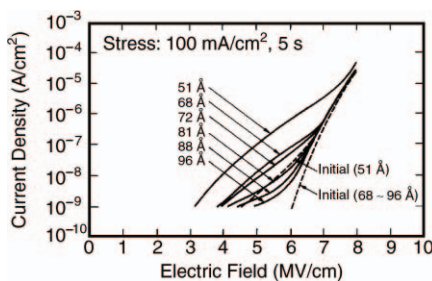


Figure 11. Stress-induced leakage current in thin oxides.¹²

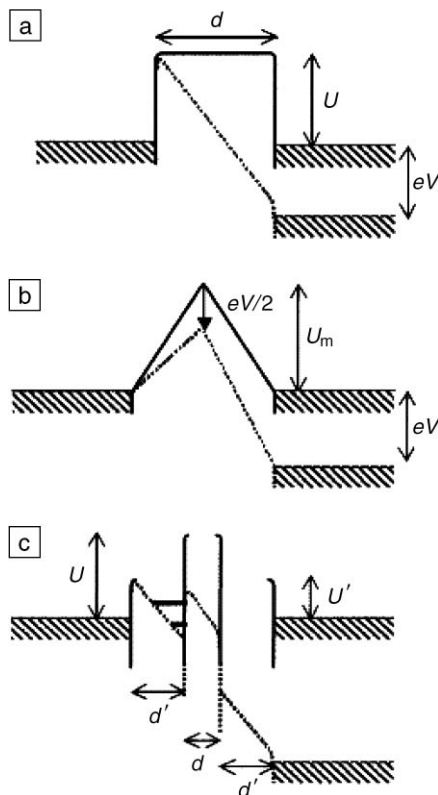


Figure 12. (a) Classical single barrier shows that tunneling is modulated by the barrier width. With barrier engineering, it is possible to have (b) a lower barrier under high field and (c) a higher barrier under low field.¹⁴ U = energy barrier height; d = distance.

barrier height reduces under the high-field condition when charge transport is desired, but increases again when a low field is present, reducing the low-field leakage current and improving retention.

Summary

Flash memory devices are now pushing into the sub-100-nm lithography regime. Device scaling is becoming challenging due to the high electric fields required for the programming and erase operations and the stringent leakage requirements for long-term charge storage. Overcoming these limitations will require innovations in cell structure and device materials. Three-dimensional structures and self-alignment techniques can address the physical scaling issues. High-dielectric-constant and crested barrier (or barrier-engineered) materials can address the dielectric scaling issues. It is projected that flash scaling can progress at the current rate through at least the end of the decade (2010) using techniques that are available today or projected to be available in the near future.

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